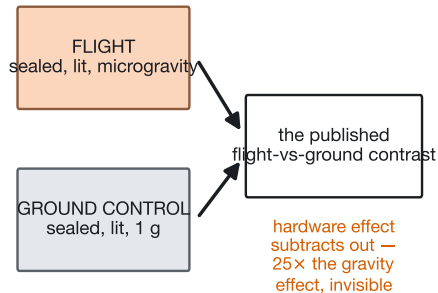


Designing the experiment that could tell hardware from gravity

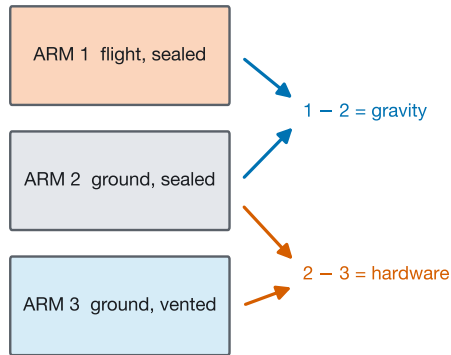
a What the standard contrast cannot see

CO₂ drawn down in BOTH arms



What survives is a persistent ~5–9 % assimilation penalty, and downstream of it a carbon-starvation signature — which is what the omics show.

b The design that separates them



Two arms confound the two effects. Three separate them — and the third arm needs no flight.

c The gap this ran into

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OSDR plant studies with a metabolome

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OSDR studies that are metabolite profiling, all mouse, human, rat or microbial

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in-canister CO₂ logs, any BRIC-class flight

A ~20-compound targeted panel on archived material would be the first plant spaceflight metabolome.